Fart A (5 x 1 = 5 marks)

(Multiple Choice Questions)

Question 1:

Assume that $\lim_{x\to 6} f(x) = -3$ and $\lim_{x\to 6} h(x) = \pi$, then the value of $\lim_{x\to 6} \frac{xh(x)}{f(x)} =$

a) -2

b) 2π

c) π

d) -2π

Question 2 :

The value of $\lim_{x\to 2^+} \frac{x-6}{2-x}$ is

a) 0

b) $-\infty$

c) $+\infty$

d) 2

Question 3 :

From the given graph answer the following:

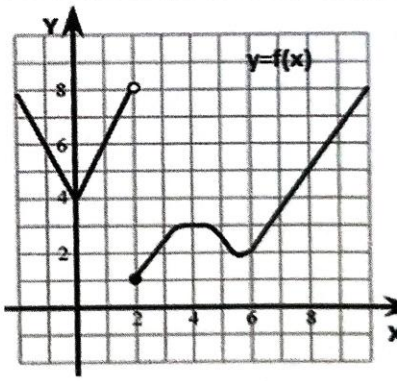
$\lim_{x\to 1^+} f(x) + \frac{f(4)}{3} =$

a) 6

b) 1

c) 7

d) -7



Question 4:

Evaluate $\lim_{x\to\infty} (1 - \frac{x-3}{x})$ is

$\frac{x-x+3}{x} = \frac{3}{x} = 0$

a) 0

b) 1

c) $+\infty$

d) $-\infty$

Question 5:

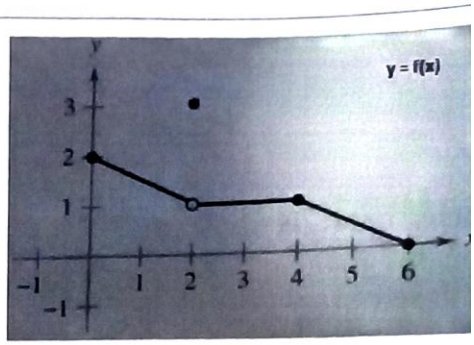
Consider the graph of $f(x)$. Which statement below is true?

a) $f(x)$ is undefined at $x = 2$

b) $f(2) = 1$

c) $\lim_{x\to 2} f(x)$ does not exist

d) $\lim_{x\to 4} f(x)$ exists



Part B (5x2=10 marks)

Question 6: **[2 Marks]**

Use the definition of derivative to find $f'(x)$ if $f(x) = \frac{1}{2x-1}$

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$

$$\lim_{h\to 0} \frac{\frac{1}{2(x+h)-1} - \frac{1}{2x-1}}{h}$$

$$\lim_{h\to 0} \frac{\frac{2x-1-(2(x+h)-1)}{(2(x+h)-1)(2x-1)}}{h}$$

$$\lim_{h\to 0} \frac{\frac{2x-1-2x-2h+1}{(2(x+h)-1)(2x-1)}}{h}$$

$$\lim_{h\to 0} \frac{-2\cancel{h}}{\cancel{h}(2(x+h)-1)(2x-1)}$$

$$\lim_{h\to 0} \frac{-2}{(2(x+h)-1)(2x-1)}$$

$$= \frac{-2}{(2x-1)(2x-1)} = \frac{-2}{(2x-1)^2}$$

2

Question 7 **[2 Marks]**

Find the vertical and horizontal asymptotes of $f(x) = \frac{(2x-1)(x+2)}{4-x^2}$

$$\frac{-2(x-1)(x+2)}{(2+x)(2-x)}$$

$$\frac{-(2x-1)}{2-x}$$

vertical:

$4 - x^2 = 0$

$-x^2 = -4$

$x^2 = 4$

$x = \pm 2$

Horizontal:

$\frac{ax^m}{bx^n}$, $n = m$

$\therefore y = \frac{a}{b} = \frac{-2}{1} = -2$

2

Question 8: **[2 Marks]**

Let $s(t) = t^3 - 2t^2 + 4t - 5$ represents the position of a moving particle, where s in feet and t in seconds.

i) Find the velocity when the acceleration is $8\ m/sec^2$.

ii) Is the object speeding up or slowing down at $t = 1\ sec$?

$s'(t) = v(t) = 3t^2 - 4t + 4$

$a(t) = v'(t) = 6t - 4$

$a(t) = 8$

$6t - 4 = 8$

$6t = 12$

$t = 2$ seconds

$v(2) = 3(2)^2 - 4(2) + 4$

i) $\rightarrow$ $= 8$ m/sec

ii)

$v(1) = 3(1)^2 - 4(1) + 4$

$= 3$ m/sec

$a(1) = 6(1) - 4$

$= 2$ m/sec^2

Same sign (+)

$\therefore$ Speeding up

2

Question 9: **[2 Marks]**

Find the values of the constants 'a' and 'k' that makes the following function continuous at $x = -4$

$$f(x) = \begin{cases} k + 3x \ ; & x < -4 \\ 2 - x^2 \ ; & x = -4 \\ -ax - k \ ; & x > -4 \end{cases}$$

2

$\lim_{x \to -4^-} K + 3x = K - 12$

$\lim_{x \to -4^+} -ax - K = -a(-4) - K$

$= 4a - K$

$K - 12 = 4a - K$

$(-2) - 12 = 4a - (-2)$

$-14 = 4a + 2$

$-16 = 4a$

$a = \frac{-16}{4}$

$a = -4$

$\lim_{x \to -4} = f(-4)$

$K - 12 = -14$

$K = -2$

$f(-4) = 2 - x^2$

$= 2 - (-4)^2$

$= -14$

$K = -2$

$a = -4$

Question 10 **Answer one of the following questions** **[2 Marks]**

a) Assume that $g(x)$ and $u(x)$ are differentiable at $x = 0$, such that $g(0) = -2$, $g'(0) = 1$, $u(0) = 4$ and $u'(0) = 6$, find $\frac{d}{dx}\left[\frac{x}{g(x)-u(x)}\right]$ at $x = 0$

OR

b) Assume that $g(x)$ and $u(x)$ are differentiable at $x = 0$, such that $g(0) = -2$, $g'(0) = 1$, $u(0) = 4$ and $u'(0) = 6$, find $\frac{d}{dx}[2u(x) - xg(x)]$ at $x = 0$

$2(u'(0) - [0(g'(0)) + 1.g(0)]$

$2(6) - 0(1)$

$= 12 + 2$

$= 14$

$2(u(0))^{x} \times 0(g'(x)) - 2u'(0) \times 0g(x)$

$2(4) \times 0 - 12 \times 0$

$= 0$

Part C (1x5=5 marks)

Question 11: **[2 Marks]**

a) **Answer one of the following questions**

i) Find the equation of the tangent line to the curve $y = x\,lnx$ at $x = e$

OR

ii) Find the equation of the tangent line to the curve $y = \frac{lnx}{x}$ at $x = e$

$y' = \frac{d}{dx}\left(\frac{\ln(e)}{e}\right) = 0 \rightarrow m$

$y = \frac{\ln e}{e} = 0.367$

$y = \frac{\ln e}{e} = \frac{1}{e}$

$y - y_1 = m(x - x_1)$

$y - (0.367) = 0\,(x - e)$

$y - 0.367 = 0$

$y = 0.367$

1.25

b) **Answer one of the following questions** **[1.5+1.5 =3 Marks]**

i) Find the derivative of:

a) $y = xsin^{-1}x - \log_3 x$

b) $y = 5^x ln2 + secx$

OR

ii) Find the derivative of

a) $y = xtan^{-1}x - 7^x$

b) $y = e^2x^\pi + cscx$

a) $y' = \frac{x \cdot 1}{1+x^2} - 7\ln(x)$

$1 \cdot \tan^{-1}x -$

b) $\frac{d}{dx}(\csc x) = -\csc x \cot x$

$y' = -\csc x \cot x$

$e^2 \pi x^{\pi - 1}$

$0.25 + 0.50 = 0.75$